

## UC5: Auction Auto-Close (System-Triggered)

**Primary Actor:** System (Timer Service)

### Stakeholders and Interests:

- *Seller*: wants a definitive, fair closing moment and to be notified of the outcome
- *Winning Bidder*: wants immediate confirmation and a clear path to pay
- *Other Bidders*: want to know they didn't win so they can move on
- *System*: must close exactly once, with no race between "closing" and a last-moment valid bid

### Preconditions:

- Auction end time has been reached (including any soft-close extensions from UC1)
- Auction status is currently "Active"

### Postconditions (success):

- Auction status changes to "Closed"
- If reserve was met (or no reserve was set) and at least one bid exists: highest bid is recorded as the winning bid
- If reserve was not met, or no bids were placed: item is marked "Unsold"
- All relevant notifications are sent (winner, seller, losing bidders)
- Closure event is logged in the audit trail

### Main Success Scenario:

1. System's timer service detects that the current time has reached the auction's end time.
2. System checks whether any bid was placed within the last soft-close window (per UC1) that would extend the timer; if none, proceeds.
3. System locks the auction from accepting further bids (status → "Closed").
4. System determines the highest valid bid.
5. System checks whether the reserve price (if any) was met.
6. System marks the highest bidder as the winner.
7. System sends the winner a "Congratulations, you won!" notification (in-app + email) with final price, item details, and a "Pay Now" button.
8. System sends the seller a notification with the final sale price and winning bidder's contact info.
9. System sends losing bidders a notification that they were outbid, with suggestions for similar items.
10. System logs the closure event (final price, winner, timestamp) in the audit trail.

### Extensions (Alternate Flows):

- **2a. A qualifying bid arrives just before the checked end time (soft-close race):**
  1. System re-evaluates: if the bid's timestamp is strictly before the (possibly already-extended) end time, it is accepted per UC1, and the end time is extended by 5 minutes.
  2. Closure is deferred; system re-checks at the new end time (back to step 1).
- **4a. No bids were ever placed:**
  1. System marks the item as "Unsold — No Bids."
  2. System notifies the seller with an option to relist.
  3. Use case ends (skip winner-related steps).
- **5a. Reserve price was set but not met:**
  1. System marks the item as "Unsold — Reserve Not Met," even though bids exist.
  2. System notifies the seller of the final (unsuccessful) highest bid and offers a relist option, possibly suggesting a lower starting price.
  3. Use case ends (no winner notification is sent).

### **Special Requirements:**

- Closure must be atomic with respect to bid acceptance — the system must never accept a bid *after* marking the auction "Closed," and must never close the auction *while* a still-valid, in-flight bid from the soft-close window is being processed. This is the core concurrency invariant for your TLA+ model.
- Auto-close should be idempotent: if the timer service fires the closure check more than once due to retries, the auction must not be closed or re-notified twice.